FURBYBOARD v1.0 · 2026-10

ESP32-S3-WROOM-1-N16R8 · BQ24073 · TPS63802 · TPS61089 + DRV8231A · MAX98357A · cámara DVP · TCA9535. Arquitectura: ../PCB-suelo/ARQUITECTURA.md

USB-C y protección

Archivo: hojas/usb.kicad_sch

Carga, batería y 3,3 V

Archivo: hojas/alimentacion.kicad_sch

ESP32-S3-WROOM-1

Archivo: hojas/mcu.kicad_sch

Elevador 6 V y puente H

Archivo: hojas/motor.kicad_sch

Amplificador y micro

Archivo: hojas/audio.kicad_sch

Cámara DVP

Archivo: hojas/camara.kicad_sch

Expansor y sensores del Furby

Archivo: hojas/sensores.kicad_sch

LED, puntos de test y J_EXP

Archivo: hojas/led_tp.kicad_sch

Generado por scripts/gen_sch.py desde scripts/design.py: no editar a mano

FurbyGPT

Sheet: /

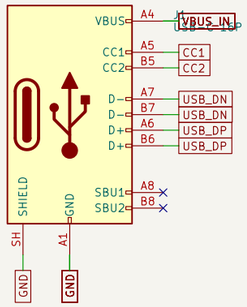
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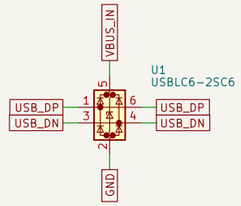
Title: FurbyBoard v1.0 · Raíz

Size: A3	Date: 2026-10-24	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 1/9

USB-C Y PROTECCIÓN

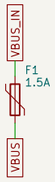
USB-C 16 pines HRO TYPE-C-31-M-13C (C2848620): D+/D- al USB nativo del ESP32-S3 (GPIO19/20).
Rd de 5,1 kΩ en CC1 y CC2: cualquier fuente USB-C da 5 V. ESD USBLC6-2SC6 junto al conector.
PTC de 1,5 A antes del cargador.

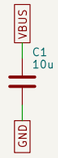




















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FurbyGPT

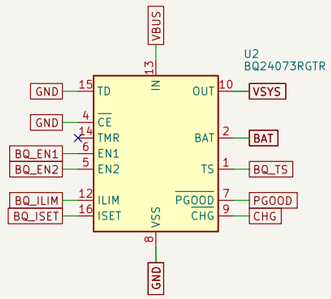
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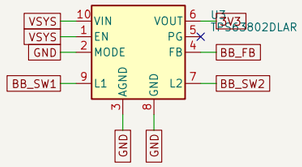
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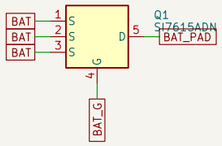
Size: A3	Date: 2026-10-24	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 2/9

CARGA, BATERÍA Y 3,3 V

BQ24073: ICHG = 890/1,8k = 0,494 A; ILIM 1610/1,2k = 1,34 A con EN2=1 y EN1=0.
Al arrancar EN1=1, EN2=0: USB500 (475 mA). OUT = 4,4 V con USB y VBAT − 50 mV sin USB.
Q1 SI7615ADN: protección de polaridad con la puerta a GND. Con la batería al revés no llega
tensión negativa a nada; con USB, el BQ se queda en detección de cortocircuito (4-11 mA).
TPS63802: 3,3 V = 0,5 × (1 + 510k/91k). LED CHG y PG desde VBUS: no gastan batería.

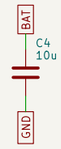






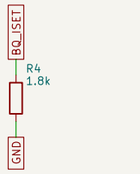


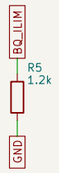


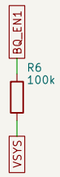


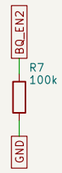


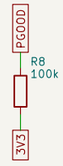


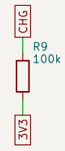










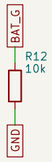




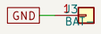


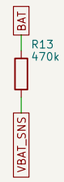








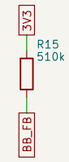


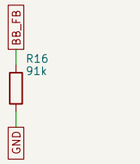
























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FurbyGPT

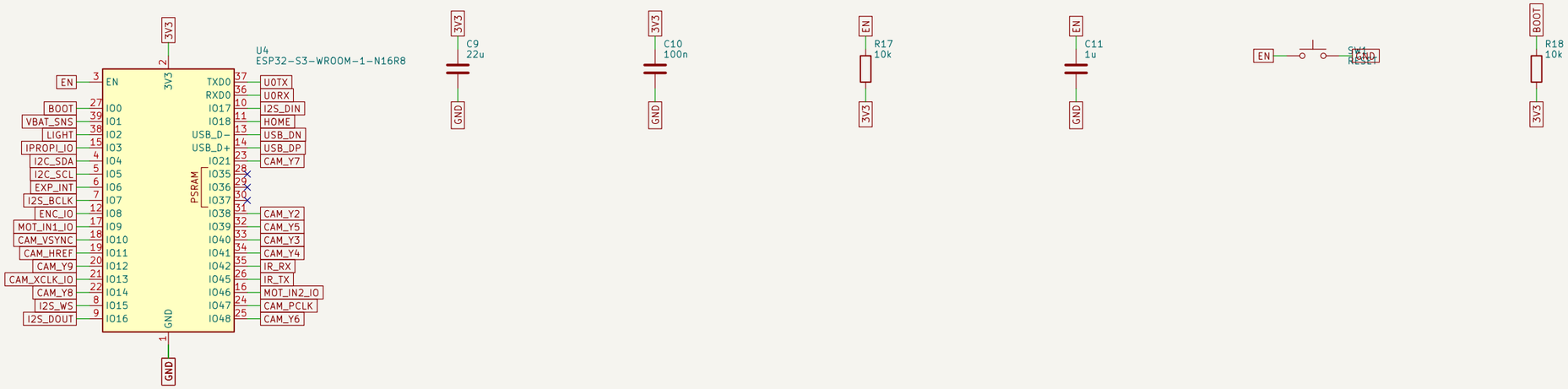
Sheet: /Carga, batería y 3,3 V/
File: alimentacion.kicad_sch

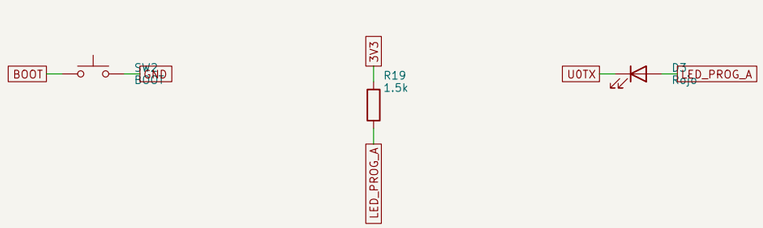
Title: FurbyBoard v1.0 · Carga, batería y 3,3 V

Size: A3	Date: 2026-10-24	**Rev: v1.0**
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ESP32-S3-WROOM-1

ESP32-S3-WROOM-1-N16R8: GPIO35-37 ocupados por la PSRAM octal.
Strapping: GPIO0 (BOOT, 10 kΩ a 3V3), GPIO45 y GPIO46 a 0 al arrancar (pull-downs en sus hojas),
GPIO3 (JTAG) ignorado sin eFuse. EN con 10 kΩ y 1 µF. LED PROG en U0TXD (log del ROM).





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FurbyGPT

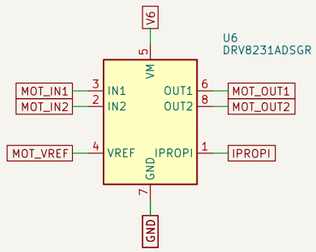
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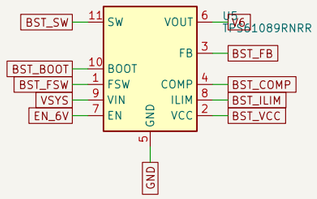
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Size: A3	Date: 2026-10-24	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 4/9

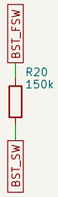
ELEVADOR 6 V Y PUENTE H

TPS61089: 5,94 V = 1,212 × (1 + 390k/100k); ILIM 3,8 A; fSW ≈ 0,96 MHz; EN desde el expansor.
DRV8231A: ITRIP = VREF / (1,5k × 1500 µA/A) = 1,0 A con VREF = 2,245 V (R26 ajustable).
PWM a 20 kHz en IN1/IN2 (el DRV se duerme si IN1 = IN2 = 0 más de 0,8 ms).
JP1-JP3: cortar para liberar GPIO9, GPIO46 y GPIO3.

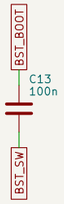




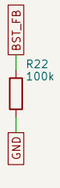


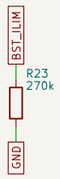


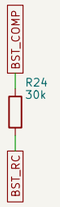


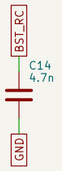


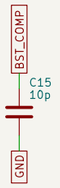


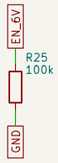


















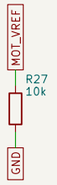




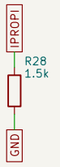


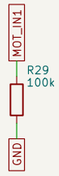


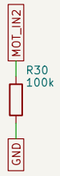










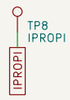












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FurbyGPT

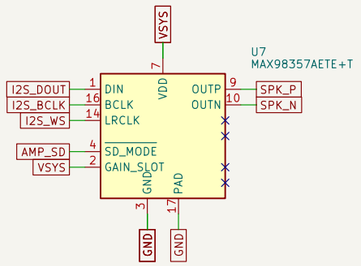
Sheet: /Elevador 6 V y puente H/
File: motor.kicad_sch

Title: FurbyBoard v1.0 · Elevador 6 V y puente H

Size: A3	Date: 2026-10-24	**Rev: v1.0**
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AMPLIFICADOR Y MICRO

MAX98357A en VSYS, GAIN_SLOT a VDD (6 dB), altavoz de 4 Ω y 3 W: 0,9-1,9 W según VSYS.
VDD: C26 (100 nF) y C41 (10 µF) junto a los pines 7/8; C24/C25 (22 µF) al otro lado del puente de VSYS.
I2S full duplex a 16 kHz: BCLK y WS compartidos con el micro ICS-43434 (placa aparte, pads J5).

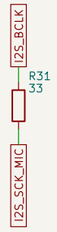


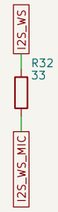


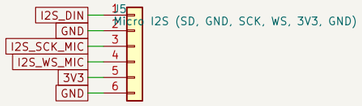


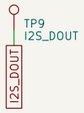












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FurbyGPT

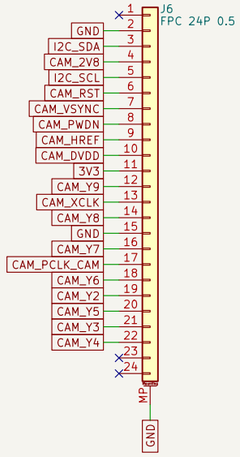
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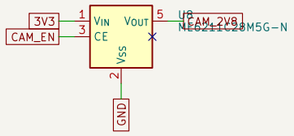
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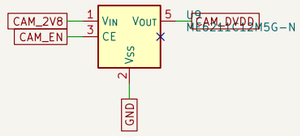
Size: A3	Date: 2026-10-24	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 6/9

CÁMARA DVP

FPC 24p 0.5 con el pinout de facto de los módulos OV2640/OV3660/OV5640 para ESP32 (verificar).
AVDD 2,8 V (U8) y DVDD 1,2 V (U9; 1,5 V con ME6211C15 para OV3660/OV5640), EN desde el expansor.
DOVDD = 3V3 siempre: el SCCB comparte el I2C del expansor. PWDN a 1 al arrancar.

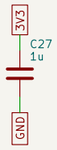




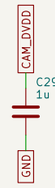




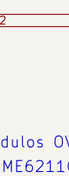


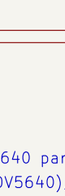


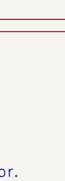


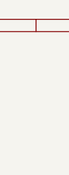












Generado por scripts/gen_sch.py desde scripts/design.py: no editar a mano
FurbyGPT

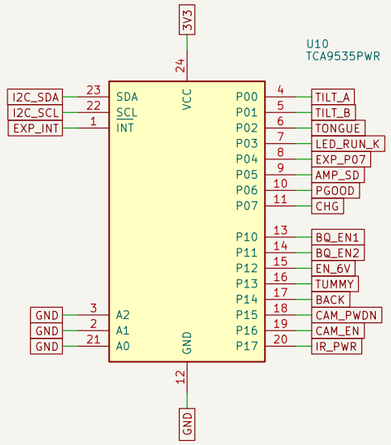
Sheet: /Cámara DVP/
File: camara.kicad_sch

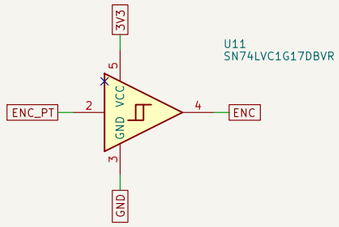
Title: FurbyBoard v1.0 · Cámara DVP

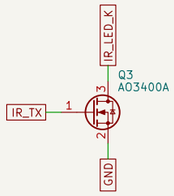
Size: A3	Date: 2026-10-24	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 7/9

EXPANSOR Y SENSORES DEL FURBY

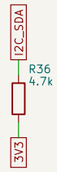
TCA9535 en 0x20, mismo I2C que el SCCB de la cámara; INT a GPIO6 (despierta del sueño).
Pull-ups: barriga, espalda y lengua 47 kΩ; inclinación 220 kΩ; leva 100 kΩ; 1 kΩ en serie.
Encoder: LED IR desde V6 por 470 Ω (luce con EN_6V) y fototransistor al Schmitt; JP4 libera GPIO8.
IR: emisor con AO3400A en GPIO45 (pull-down por el strapping); receptor alimentado por P17.

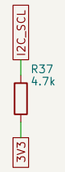


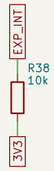












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FurbyGPT

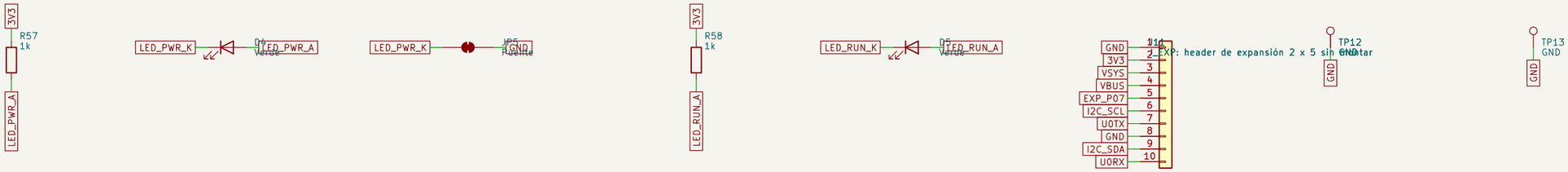
Sheet: /Expansor y sensores del Furby/
File: sensores.kicad_sch

Title: FurbyBoard v1.0 · Expansor y sensores del Furby

Size: A3	Date: 2026-10-24	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 8/9

LED, PUNTOS DE TEST Y J_EXP

LED PWR (JP5 para apagarlo), LED RUN en P03 del expansor.
J_EXP: 2 × 5 sin montar: 1 GND, 2 3V3, 3 VSYS, 4 VBUS, 5 P04, 6 SCL, 7 U0TX, 8 GND, 9 SDA, 10 U0RX.



Generado por scripts/gen_sch.py desde scripts/design.py: no editar a mano
FurbyGPT

Sheet: /LED, puntos de test y J_EXP/
File: led_tp.kicad_sch

Title: FurbyBoard v1.0 · LED, puntos de test y J_EXP

Size: A3	Date: 2026-10-24	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 9/9